

SYLLABUS & SCHEME OF EXAMINATION

**DIPLOMA IN MEDICAL LABORATORY
TECHNOLOGY (DMLT)**



SESSION 2021-22 & ONWARDS

REGULATION

REGULATION FOR VARIOUS DIPLOMA COURSES (2 YEARS) IN PARAMEDICAL SUBJECTS

The objective of the courses

1. The training of the candidate registering for two year diploma course is aimed to carry out all routine diagnostic / therapeutic tests on modern hospital laboratory
2. The Diploma course is of two years
3. Eligibility for admission in Para medical Diploma course
 - a) The candidate must have passed 10+2 with at least 45% marks in aggregate of Physics Chemistry and Biology (P C B) for DMLT will be eligible
4. In case of SC/ST/OBC candidates, 5% marks relaxation will be given for the admission in above said courses
5. Number of seats will be 50.
6. The minimum age for admission shall be **17 years** on 31 December of the Academic year of admission
7. The selection of candidate will be on the basis of the entrance examination or in the absence of entrance exam, purely on merit list prepared on the basis of marks obtained in qualifying examination
8. The admissions of students into diploma programs should be completed by **31st July**. There shall be **minimum 240 teaching days** in one academic year.
9. **Attendance of students:** The student should have 75% attendance in theory and 75% attendance in practical to be eligible to appear in university examination.

However, only Dean/ Principal of the college are empowered to condone 10% of attendance on valid grounds. Monthly attendance of students has to be sent to the university every month.
10. **Total Course Duration:** A student has to complete the course that he/she has joined with in the double the duration of the actual time taken for the completion of the course i.e. - Four Years for Diploma Course.
11. **Medium of instruction and examination:** In case of diploma course the medium of instruction will be English and the medium of examination will be English.
12. There will be one Internal and University Examinations in one academic session.
13. **Passing Marks of Examination:**
 - The passing marks of examination would be 50% for each subject and also in total marks obtained for both Diploma and Degree courses. The candidate has to pass in theory and practical examination separately.

For theory paper the internal assessment marks, viva-voce and theory examination marks will be counted.

 - The candidate has to pass separately in each subject in internal assessment examination (with 50% marks) in order to be eligible to appear in university examinations.
 - Regarding Grace Marks it was decided that total weightage of grace marks would be 5 (five), and that grace marks can be split between the subjects. There will not be any grace marks for the Practical examination.
 - Grace marks will not be added to total marks of the candidate. In Supplementary examination also similar pattern of grace marks will be followed.

14. Re-totaling & Re-valuation (Review) both should be allowed as per the university rules

15. Other rules regarding conduct of examination will be as per the university

REGULATION/notifications

16. Permission will be given to start the courses after the permission from M P Paramedical council (Department of Medical education, Government of M. P.)

17. Syllabus and scheme of examination shall be decided by the board of studies and Academic council of the university from time to time as per guidelines from M P Paramedical council

18. In order to be an examiner a faculty member should have minimum three years of teaching experience and not below the rank of the Associate Professor/ Assistant Professor.

19. The examination papers (Theory) will be evaluated centrally at the university

20. The examiner who evaluates the theory copy preferably be appointed as the external for practical examinations for the said course.

21. Diploma students will be eligible for supplementary examination.



FACULTY OF PARAMEDICAL SCIENCE

BHABHA UNIVERSITY, BHOPAL

(Approved by M.P. Paramedical Council)

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SCHEME OF MARKS FOR DMLT 1ST YEAR

		Internal Assessment	University Examination	
Sub. Code	Subjects	Theory	Theory, Practical & Viva Voce	Total
DMLT101	Human Anatomy & Physiology	50 Marks	150 Marks	200
DMLT102	Hematology, Clinical Pathology & Parasitology	50 Marks	150 Marks	200
DMLT103	Clinical Biochemistry	50 Marks	150 Marks	200
Total				600

SCHEME OF MARKS FOR DMLT 2nd YEAR

		Internal Assessment	University Examination	
Sub. Code	Subjects	Theory	Theory, Practical & Viva Voce	Total
DMLT201	Blood Banking & Immuno-Serology	50 Marks	150 Marks	200
DMLT202	Medical Microbiology	50 Marks	150 Marks	200
DMLT203	General Pathology, Histopathology & Cytopathology	50 Marks	150 Marks	200
Total				600

NOTE: Internal Assessment marks will be added in theory marks; candidate have to get min. 50% marks i.e.- 75 marks in theory and internal assessment collectively for passing the examination and in practical he/she should get 50% marks i.e.-25 marks to get pass.

All theory papers will of maximum 100 marks and time duration 3Hrs.

Pattern of Theory Examination will be as under:-

Type of Questions	Number of Questions	Marks for each Question	Total Marks
Very Short Questions	10	2	20
Short Questions	5	10	50
Long Question	2	15	30
			100

SYLLABUS & SCHEME OF EXAMINATION FOR DMLT 1ST YEAR

S. No.	Subject Code	Subject	Theory				Practical		Total Marks
			UE	IA	Total	Minimum Marks	Practical & Viva Voce	Minimum Marks	
1	DMLT101	Human Anatomy & Physiology	100	50	150	75	50	25	200
2	DMLT102	Hematology, Clinical Pathology & Parasitology	100	50	150	75	50	25	200
3	DMLT103	Clinical Biochemistry	100	50	150	75	50	25	200
Grand Total									600

Paper: I Human Anatomy & Physiology

Course Code: DMLT101

Course Contents: Theory

An Introduction to the Human Body: Definition and divisions of anatomy. Anatomical terms related to positions, planes and movements.

Cellular & Tissue Level of the Organization: Cell- Definition, Structure, Function and cell division. Tissue- Definition, classification, structure and function of Epithelial Tissue, Connective Tissue, Muscular Tissue, Nervous Tissue and Glands.

Musculo-Skeletal System: Bone- Definition, structure, types and function of bones. Joints- Definition, classification and function of joints. Muscles- Definition, classification, structure and function.

Digestive System: Structure and function of esophagus, stomach, small and large intestine, liver pancreas. Structure and function of accessory organ of digestive system. Composition and function of saliva, gastric juice, bile juice and pancreatic juice. Digestion and absorption of carbohydrate, protein and lipids.

Renal System: Structure and function of Kidneys, ureters, urinary bladder and urethra. Nephron- structure and function. Mechanism of urine formation.

Respiratory System: Structure and function of Nose, Pharynx, larynx, trachea, lungs, . Mechanism of respiration. Exchange of gases. Pulmonary function test.

Circulatory and Lymphatic System: Heart- structure and function. Heart rate, cardiac cycle, blood pressure, pulse rate, etc. Pulmonary and systemic circulation. Blood vessels- Aorta, artery, vein, etc. difference between artery and vein. Blood and Blood group system. \

Lymphatic System: Lymph node, spleen and Reticuloendothelial system.

Nervous System: Structure and function of Central nervous system- Brain and spinal cord, Peripheral nervous system and autonomic nervous system. Neuron- structure and function.

Endocrine System: Structure, functions and hormones of pituitary, thyroid, parathyroid, pineal, thymus, pancreatic, adrenal gland, etc.

Special Sense: Structure and functions of skin, eye, ear, nose and tongue.

Reproductive System: Male reproductive organs- Structure and function of testis, Duct system, prostate and urethra. Spermatogenesis, semen. **Female reproductive organs-** Structure and function of Uterus, ovary vagina, cervix. Menstrual cycle, Oogenesis, Fertilization. Mammary gland- Structure and function. Fertility and Infertility. Fertility Control

Course Contents: Practical

Body Temperature, pulse rate, Blood pressure, ABO & Rh Grouping

Recommended Books

B.D. Chaurasia's General Anatomy- Krishna Garg

Ross & Wilson Anatomy & Physiology- Waugh Grant

Applied Anatomy & Physiology for Nursing Students- Krishna Garg

Essential of Medical Physiology- K. Sembulingam

Paper- II Hematology, Clinical Pathology & Parasitology

Course Code: DMLT102

Course Contents: Theory

Hematology

Introduction to hematology. Anticoagulants, Diluting fluids, Romanowsky stain and other reagents.
Collection and preservation of blood.
Basic instruments- Microscope, Hemoglobinometer, Hemocytometer, RBC pipette, WBC pipette, Hb pipette, ESR pipette and tube, etc.
Blood- Definition, properties, composition and function.
Haematopoiesis- Erythropoeisis, Leukopoeisis and Thrombopoeisis
Hemoglobin- Structure, function and its type.
Anemia- Definition, classification and Laboratory diagnosis.
Morphology of blood cells- Normal and abnormal.
Normal and absolute value in Hematology.
Leukemia- Definition, classification and Laboratory diagnosis.
Bone marrow aspiration and biopsy.
Blood coagulation factor and clotting mechanism.
Bleeding and Coagulation disorders.
Quality control and Automation in hematology.

Clinical Pathology

Introduction- Instruments and reagents used in Clinical Pathology
Urinalysis and Stool Examination- Physical, chemical and microscopic examination.
Cerebrospinal Fluid, Semen and other body fluids examination.

Parasitology

Protozoology: Amoeba (Entamoeba), Flagellates (Giardia, Trichomonas, Leishmania, Trypanosoma), Sporozoa (Plasmodium, Toxoplasma), Ciliate (Balantidium).
Helminthology: Cestodes (Taenia, Hymenolepis, Echinococcus), Trematodes (Fasciola, Schistosoma)
Nematodes (Ascaris, Trichuris, Ancylostoma, Strongyloides, Enterobius, Wuchereria)

Course Contents: Practical

Erythrocyte count, PCV, Red cell indices, ESR, Leukocyte count, Platelet count, Reticulocyte count.

BT, CT, PT, PTT, APTT, L.E Cell preparation.

Peripheral blood and Bone marrow smear- Preparation and staining methods

Physical, chemical examination of urine, stool, etc

Microscopic examination of stool for parasitic detection

Recommended Books

Text and Practical Hematology for MBBS- Tejindar Singh

Hematology for Students and Practitioners- Ramnik Sood

Essentials of Clinical Pathology- S. M. Kawthalkar

Complete Microbiology for MBBS- C.P. Baveja

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Wintrobe's Clinical Hematology- John P. Greer

Paper- III Clinical Biochemistry

Course Code: DMLT103

Course Contents: Theory

Introduction- Units of measurement. Biochemical instruments and their care and maintenance. Laboratory chemical and reagent and their preparation.

Carbohydrate- Definition, classification, properties and function.

Protein, Plasma protein and Amino acid- Definition, classification, properties and function.

Lipid- Definition, classification, properties and function of lipids.

Enzyme- Definition, classification, properties and function of enzymes.

Water and Mineral metabolism. Acid- Base balance. Vitamins and Nutrition.

Liver function test, Kidney function test, Thyroid profile, Cardiac profile, etc

Glucose tolerance test, Urea and Creatinine clearance.

Cancer and tumor marker

Polymerase chain reaction, Radioimmunoassay, ELISA, Electrophoresis, Chromatography

Quality control and Automation in clinical biochemistry.

Course Contents: Practical

Glucose, Total protein, Albumin, Globulin, A/G ratio,

Cholesterol, Triglyceride,

Urea, Creatinine, Uric acid,

SGPT, SGOT, Bilirubin, etc

Recommended Books

Textbook of Medical Biochemistry for Medical Students- DM Vasudevan

Biochemistry- Satyanarayana

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Tietz Fundamental of Clinical Chemistry and Molecular Diagnostics- Nader Rifai

Varley's Practical Clinical Biochemistry- Harold Varley

SYLLABUS & SCHEME OF EXAMINATION FOR DMLT 2ND YEAR

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Grand Total									600

Paper- I Blood Banking & Immuno-Serology

Course Code: DMLT201

Course Contents: Theory

Blood Banking

Anticoagulant and Blood collection

ABO System and Rh System.

Blood grouping techniques- Forward and reverse grouping.

Blood donation- Blood donor and their types.

Criteria for selection and rejection of donor.

Testing of blood donor- Coomb's test, Screening of transfusion transmitted disease, etc

Compatibility testing- Cross matching

Hemolytic disease of newborn

Blood transfusion reaction- Types and their investigation.

Blood components and their preparation- Whole blood, Blood Components and Blood Derivatives.

Aphaeresis

Immuno-Serology

Immunity. Antigen and Antibody (Immunoglobulin)

Antigen-antibody reaction

Compliment system, Hypersensitivity Reaction, Autoimmunity,

Immunodeficiency disease, Immunoprophylaxis

Course Contents: Practical

ABO blood grouping and Rh typing. Preparation of red cell suspension.

Coomb's Test, Cross Match, D^u test

Transfusion transmitted tests- HBsAg, HIV, HCV, etc

Recommended Books

Essential of Blood Banking and Transfusion Medicine- Ganga S. Pilli

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Modern Blood Banking and Transfusion Medicine- Denise M. Harmening

Paper- II Medical Microbiology

Course Code: DMLT202

Course Contents: Theory

General Microbiology

Introduction and History. Contribution of some important Scientists.

Microscopy and Staining Method

Morphology of Bacteria. Physical and Nutritional requirement for growth of bacteria.

Bacterial growth curves.

Culture Media and Culture Method, Identification of bacteria

Sterilization, Disinfection and Antiseptic.

Infection- Definition, types, source and mode of infection

Collection and transportation of microbiological specimen

Bacteriology

Staphylococcus, Streptococcus, Pneumococcus, Neisseria, Corynebacterium, Mycobacterium, Nonsporing anaerobes, Spirochaetes, Enterobacteriaceae (Escherichia, Klebsiella, etc) Proteus, Shigella, Salmonella, Vibrio, Pseudomonas, Haemophilus, Chlamydia, etc

Virology

DNA viruses- Poxviruses, Herpesviruses, Adenoviruses, Parvoviruses, Papovaviruses,

RNA viruses- Picornaviruses, Orthomyxoviruses, Paramyxoviruses, Arboviruses, Rhabdoviruses,

Human immunodeficiency virus, Coronaviruses, Filoviruses, Rubivirus, etc

Miscellaneous viruses- Hepatitis virus, Oncogenic virus.

Mycology

Superficial mycoses (Dermatophytosis, Tinea, etc), Subcutaneous mycoses (Mycetoma, etc), Systemic mycoses (Histoplasmosis, etc), Opportunistic mycoses (Candida, Cryptococcosis, Aspergillosis, etc)

Clinical & Applied Microbiology

Normal microbial flora of the human body, Urinary tract infection,

Diarrheal diseases, Meningitis, Blood stream infection, Respiratory tract infection, etc

Hospital acquired infection- Definition and types.

Laboratory acquired infection.

Biomedical waste management.

Quality control in microbiology.

Zoonoses .

Course Contents: Practical

Autoclave and Hot air oven

Gram stain, ZN Stain and Albert stain.

Hanging drop preparation.

ASO, CRP, VDRL, Widal, RA Factor,

Pregnancy test, other serological test.

Preparation of different culture media.

Antimicrobial sensitivity test

Identification of bacteria by biochemical tests.

India ink preparation, KOH preparation,

Lactophenol cotton blue staining,

Recommended Books

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Complete Microbiology for MBBS- C.P. Baveja

Ananthanarayan and Paniker's Textbook of Microbiology- Reba Kanungo

Bailey & Scott's Diagnostic Microbiology- Patricia M. Tille

Paper- III General Pathology, Histopathology & Cytopathology

Course Code: DMLT203

Course Contents: Theory

General Pathology

Cell injury, Inflammation and Healing.

Neoplasm: Definition and classification.

Difference between benign and malignant tumor.

Histopathology

Introduction, Basic instrument- Tissue processor, Wax bath, Water bath, Microtome, Microtome knife, Hoon, Strop, L-Mold, etc. Cyto-Histopathological reagents.

Microtome knife and knife sharpening.

Fixatives and Fixation of tissue

Decalcifying fluids and Decalcification of tissue.

Tissue processing.

Embedding media and Embedding.

Microtomy and section cutting.

Section cutting faults and their cause and remedies.

Slide preparation and Staining procedure and Mounting

Staining for microorganisms.

Frozen Section and Cryostat

Autopsy and Museum Technique

Cytopathology

Introduction to Exfoliative cytology especially for gynecological specimen.

Collection, slide preparation and staining procedure in cytology.

Fine needle aspiration cytology

Course Contents: Practical

Microtome, Tissue processing,

FNAC

H & E Stain, Pap smear, etc

Recommended Books

Pathology Practical- P. Chakraborty

Manual of Histological Techniques- Santosh Kumar Mandal

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Bancroft's Theory and Practice of Histological Techniques- S. Kim Suvarna